

Compute-Optimal Scaling Laws

From spectra to learning curves

Princeton 2026 Summer School in Machine Learning

Module 1 left us a model for activation covariance

At a fixed layer, let X denote the random activation vector and

$$\Sigma = \mathbb{E}[X \otimes X].$$

A useful empirical model is a long spectral tail,

$$\lambda_j(\Sigma) \asymp j^{-\alpha}.$$

This is a **linear** description of the activations: it says that variance is spread across a hierarchy of directions.

How can this hierarchy influence the loss we can reach with a finite amount of compute?

Fixed compute turns learning into an allocation problem

Suppose a model of size d is trained for r steps at batch size B :

$$C \asymp dBr.$$

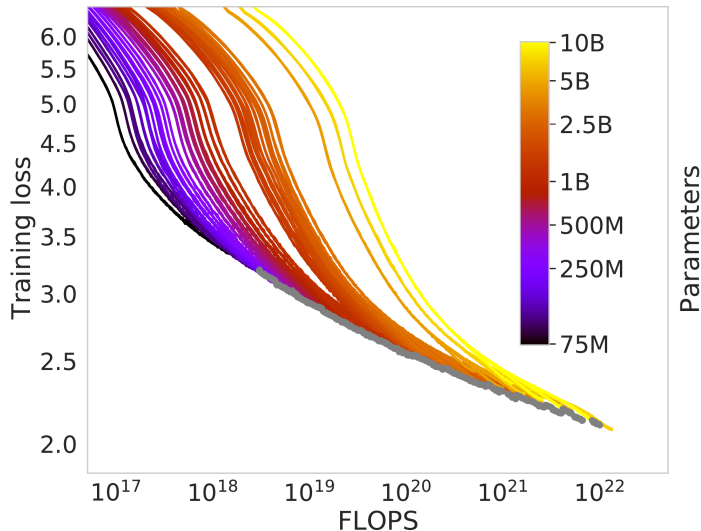
At fixed C , changing model size also changes the number of updates:

$$r \asymp \frac{C}{dB}.$$

For each compute budget, which model size achieves the lowest final loss?

The resulting curve $(d_*(C), \mathcal{L}_*(C))$ is the **compute-optimal frontier**. It will organize the lecture.

Hoffmann et al. recover the frontier from many training runs



What varies

- Models from 70M to 10B parameters
- Four training horizons for each size
- Loss recorded throughout every run

What survives

At each FLOP budget, retain the smallest observed loss. The gray curve is this lower envelope.

Compute optimality is a constrained minimization

Train a model of size d for r steps at batch size B :

$$C \asymp dBr, \quad d_{\star}(C) \in \arg \min_d \mathcal{L}\left(\frac{C}{dB}, d\right).$$

For each compute budget, an IsoFLOP experiment estimates the lower envelope $\mathcal{L}_{\star}(C)$ and its minimizing model size.

The allocation law is determined by whichever loss terms are still visible at the optimum.

Square-root allocation is important but not universal

The Chinchilla experiments found an approximately even logarithmic allocation between parameters and training tokens:

$$d_{\star}(C) \asymp C^{1/2}, \quad r_{\star}(C) \asymp C^{1/2}.$$

The exponent $1/2$ is not forced by $C \asymp dr$. It must arise from the exponents of the competing loss terms.

Our goal is to derive those terms from a covariance model, then ask when their balance produces the square-root law.

Hoffmann et al. (2022); Paquette–Paquette–Xiao–Pennington (2024).

**Can a simple covariance model
reproduce this frontier?**

A solvable regression model lets us analyze the frontier

Let $Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and define

$$X_j = j^{-\alpha/2} Z_j, \quad \Sigma = \mathbb{E}[X \otimes X] = \text{diag}(j^{-\alpha}).$$

The supervised target is

$$Y = \langle X, b \rangle, \quad b_j = j^{-\beta}.$$

Data exponent α

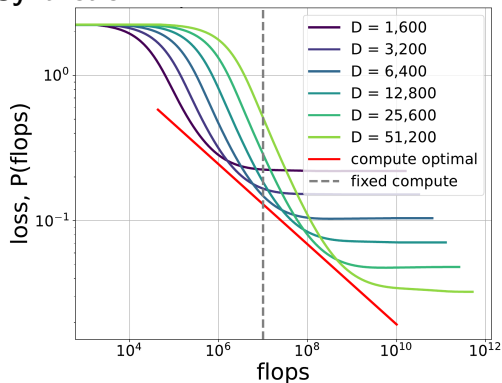
How quickly variance disappears into the tail.

Target exponent β

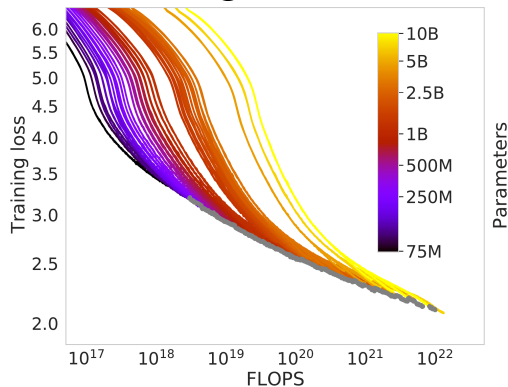
How strongly the target uses those tail directions.

The PLRF model mirrors Chinchilla's lower envelope

Synthetic PLRF



Chinchilla training runs



The gray envelope on the right is the empirical analogue of the red compute-optimal frontier on the left.

Left: course PLRF example, $\alpha = 0.8$, $\beta = 0.4$. Right: Hoffmann et al. (2022), Figure 2.^{8/37}

Random features turn ambient structure into model capacity

Use a Gaussian feature map

$$W \in \mathbb{R}^{v \times d}, \quad W_{jk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1/d), \quad \phi(X) = W^\top X.$$

The predictor $\hat{Y}_\theta = \langle W^\top X, \theta \rangle$ has population loss

$$\mathcal{L}(\theta) = \mathbb{E}[(\hat{Y}_\theta - Y)^2] = \left\| \Sigma^{1/2}(W\theta - b) \right\|_2^2.$$

d is not only a scalar capacity. The random span also distorts the population eigenbasis through which the target is learned.

Fresh mini-batches isolate optimization noise

Fresh batches remove finite-dataset reuse:

$$\theta_{r+1} = \theta_r - \gamma \hat{\nabla} \mathcal{L}_r(\theta_r).$$

coordinate	role
α	covariance geometry
β	target alignment
d	random-feature capacity
(γ, B, r)	optimization and compute

The Hessian is a deformed sample covariance

Conditioning on W , the feature-space Hessian is

$$\hat{K} = W^\top \Sigma W.$$

Its nonzero eigenvalues equal those of

$$H = \Sigma^{1/2} W W^\top \Sigma^{1/2}, \quad \text{spec}_+(\hat{K}) = \text{spec}_+(H).$$

Learning curves use $(I - \gamma \hat{K})^r$ and $(\hat{K} - zI)^{-1}$.

The optimizer sees a random sample-covariance resolvent, not Σ directly.

Random matrix lens

Isotropic sample covariance does not converge to the identity

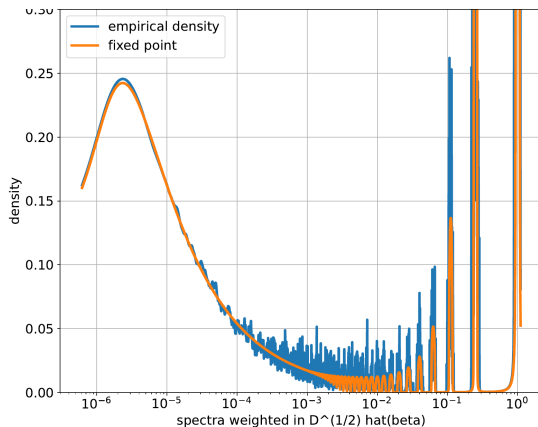
Let $G \in \mathbb{R}^{v \times d}$ have iid standard Gaussian entries and set $H = d^{-1}GG^\top$, with $v/d \rightarrow c$.

Theorem (Marchenko–Pastur)

The empirical spectrum of H converges to a deterministic law supported on $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$, with a zero atom when the matrix has a macroscopic nullspace.

The sample spectrum has order-one width even though $\mathbb{E}[H] = I$.

A scalar fixed point resolves a deformed spectrum



Course example: $\alpha = 2$, $\beta = 2$, $d = 1000$, $v = 10000$.

The resolvent turns spectral questions into analysis

Write

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_\nu), \quad W = d^{-1/2}G, \quad H = \Lambda^{1/2}WW^\top\Lambda^{1/2}, \quad R(z) = (H - zI)^{-1}.$$

For $\text{Im } z > 0$,

$$\frac{1}{\nu} \text{Tr } R(z) = \frac{1}{\nu} \sum_{\ell=1}^{\nu} \frac{1}{x_\ell - z}.$$

Contour integrals of $R(z)$ recover spectral densities, projectors, and the training propagators entering the loss.

The matrix derivative is the only calculus we need

Start from

$$RH = I + zR, \quad RH = (R\Lambda^{1/2}W)(W^\top\Lambda^{1/2}).$$

For $W \mapsto W + \varepsilon\Delta$,

$$\partial_\Delta H = \Lambda^{1/2}(\Delta W^\top + W\Delta^\top)\Lambda^{1/2}, \quad \boxed{\partial_\Delta R = -R(\partial_\Delta H)R.}$$

The factorization exposes one Gaussian matrix; the directional derivative tracks how the resolvent changes when that matrix moves.

Gaussian integration by parts can be matrix-valued

Let $\widetilde{W}$ be an independent copy of W . For a smooth compatible matrix function F ,

$$\mathbb{E}[F(W)W^\top] = \mathbb{E}[(\partial_{\widetilde{W}} F(W))\widetilde{W}^\top].$$

Take $F(W) = R(W)\Lambda^{1/2}W$, so the exposed product is $F(W)W^\top\Lambda^{1/2} = RH$. Using $\partial R = -R(\partial H)R$, the identity becomes

$$\begin{aligned}\mathbb{E}[RH] = \mathbb{E}\bigg[& R\Lambda^{1/2}\widetilde{W}\widetilde{W}^\top\Lambda^{1/2} \\ & - R\Lambda^{1/2}(\widetilde{W}W^\top + W\widetilde{W}^\top)\Lambda^{1/2}R\Lambda^{1/2}W\widetilde{W}^\top\Lambda^{1/2} \bigg].\end{aligned}$$

Two Wick contractions have different geometry

Condition on W . The purple factors are paired under $\mathbb{E}_{\widetilde{W}}$:

$$\mathbb{E}[RH] = \mathbb{E}\left[R\Lambda^{1/2}\widetilde{W}\widetilde{W}^\top\Lambda^{1/2} - R\Lambda^{1/2}(\widetilde{W}W^\top + W\widetilde{W}^\top)\Lambda^{1/2}R\Lambda^{1/2}W\widetilde{W}^\top\Lambda^{1/2}\right].$$

Coherent contraction

Incoherent contraction

$$\mathbb{E}_{\widetilde{W}}[\widetilde{W}A\widetilde{W}^\top] = \frac{1}{d} \text{Tr}(A)I.$$

$$\mathbb{E}_{\widetilde{W}}[\widetilde{W}^\top BC\widetilde{W}^\top] = \frac{1}{d} C^\top B^\top.$$

The first pairing closes a trace; the second leaves a matrix correction.

The coherent term creates the self-consistent scalar

The two contractions give the exact master equation

$$\mathbb{E}[RH] = \mathbb{E}\left[q_d(z)R\Lambda - \frac{1}{d}RHR\Lambda\right], \quad q_d(z) = 1 - \frac{1}{d} \text{Tr}(RH).$$

Using $RH = I + zR$, its j th diagonal entry is

$$\mathbb{E}[(\lambda_j q_d - z)R_{jj}] = 1 + \frac{\lambda_j}{d} \mathbb{E}[(RHR)_{jj}].$$

The coherent contraction produces q_d ; the incoherent contraction is the explicit d^{-1} correction.

Trace concentration closes the random equation

For fixed $\text{Im } z > 0$,

$$\|R(z)\| \leq \frac{1}{\text{Im } z}, \quad \text{Var}\left(\frac{1}{d} \text{Tr}(\Lambda R(z))\right) \rightarrow 0.$$

Drop the explicit d^{-1} term and factor the concentrated trace:

$$\mathcal{R}_{jj}(z) = \frac{1}{\lambda_j m(z) - z}, \quad m(z) = \left[1 + \frac{1}{d} \sum_{j=1}^v \frac{\lambda_j}{\lambda_j m(z) - z} \right]^{-1}.$$

A large random matrix has been replaced by one nonlinear scalar equation.

Dynamics: a scalar Volterra equation

SGD risk satisfies a scalar Volterra equation

Condition on the feature map, let $Hu_j = \lambda_j u_j$, and write

$$P_r = \mathbb{E}[e_r^\top H e_r], \quad q_{0,j} = \mathbb{E}[\langle e_0, u_j \rangle^2], \quad a_j = (1 - \gamma \lambda_j)^2 + \frac{\gamma^2}{B} \lambda_j^2.$$

For Gaussian mini-batches,

$$\begin{aligned} P_r &= F_r + \sum_{s=0}^{r-1} K_{r-1-s} P_s, \\ F_r &= \sum_j \lambda_j a_j^r q_{0,j}, \quad K_\ell = \frac{\gamma^2}{B} \sum_j \lambda_j^2 a_j^\ell. \end{aligned}$$

Forcing and kernel separate descent from volatility

Forcing F_r

- mean-gradient relaxation;
- target overlap with each spectral mode;
- finite-feature approximation error.

Kernel K_r

- variance injected by a mini-batch;
- propagated through later iterations;
- fed back in proportion to the current loss.

Random matrix theory makes the random F and K deterministic. The Volterra equation then propagates those spectral inputs through time.

Stability is collective, not only a top-eigenvalue condition

Mean-square stability

$$\|K\|_1 < 1,$$

$$\|K\|_1 = \frac{\gamma}{B} \sum_i q_i,$$

$$q_i = \frac{\lambda_i}{2 - \gamma(1 + 1/B)\lambda_i}.$$

Collective scale

$$r_{\text{eff}}(H) = \frac{\text{Tr}(H)}{\|H\|},$$

$$B_{\text{crit}} \asymp r_{\text{eff}}(H).$$

Below B_{crit} , larger batches permit nearly proportional step-size growth; above it, the gain saturates.

Fixed learning rate

From here on, we take γ to be fixed inside the stable regime.

A heavy-tailed memory kernel has one dominant summand

Lemma (Kesten tail reduction)

For a stable, nonnegative subexponential kernel, one large summand dominates a long convolution:

$$\frac{(K * K)_r}{K_r} \longrightarrow 2\|K\|_1.$$

Thus $P = F + K * F + K^{*2} * F + \dots$ reduces in the PLRF scaling regime to

$$\mathcal{P}_r \asymp \mathcal{F}_r + \frac{1}{\gamma B} \mathcal{K}_r.$$

Four terms become phases

The relevant spectrum is weighted by the residual

Let $Hu_\ell = x_\ell u_\ell$ and $y = \Lambda^{1/2}b$. Define

$$\hat{\mu}_{\mathcal{F}} = \sum_{\ell=1}^v |\langle u_\ell, y \rangle|^2 \delta_{x_\ell}.$$

Then the deterministic descent forcing is

$$\mathcal{F}(r) = \int \varphi_r(x) \mu_{\mathcal{F}}(dx), \quad \varphi_r(x) \approx e^{-2\gamma r x}.$$

An unweighted eigenvalue histogram is insufficient: the target overlaps decide which spectral modes contribute to the loss.

The weighted spectrum splits into peaks and feature bulk

For $\alpha > 1$, $\beta > 1/2$, and mesoscopic $d^{-\alpha} \ll c \ll 1$,

$$\mu_{\mathcal{F}}([c, 2c]) \asymp \underbrace{c^{1+(2\beta-1)/\alpha}}_{\text{population peaks}} + \underbrace{\frac{1}{d} c^{1-1/\alpha}}_{\text{feature bulk}}.$$

Equivalently, in a sufficiently weak topology that averages over the microscopic peaks,

$$d\mu_{\mathcal{F}_{\text{pp}}}(x) \asymp x^{(2\beta-1)/\alpha} dx, \quad d\mu_{\mathcal{F}_{\text{ac}}}(x) \asymp d^{-1} x^{-1/\alpha} dx.$$

Integrating the two spectral pieces gives two biases

Apply the training filter $e^{-2\gamma r x}$:

$$\begin{aligned}\mathcal{F}_{\text{pp}}(r) &\asymp \int_0^\infty e^{-2\gamma r x} x^{(2\beta-1)/\alpha} dx \asymp r^{-(1+(2\beta-1)/\alpha)}, \\ \mathcal{F}_{\text{ac}}(r, d) &\asymp \frac{1}{d} \int_0^\infty e^{-2\gamma r x} x^{-1/\alpha} dx \asymp d^{-1} r^{-1+1/\alpha}.\end{aligned}$$

Population bias is ordinary modewise relaxation. Embedding bias is created by the random-feature bulk.

The capacity floor is target mass at the zero atom

$$\mathcal{F}_0 = \lim_{\eta \downarrow 0} \eta y^\top (H + \eta I)^{-1} y, \quad d = \sum_j \frac{\lambda_j}{\lambda_j + \kappa}.$$

$$\kappa \asymp d^{-\alpha} \implies \text{spectral cutoff at } j \asymp d.$$

Splitting the target energy at $j \asymp d$ gives

$$\boxed{\mathcal{F}_0(d) \asymp d^{-\alpha + \max\{0, 1-2\beta\}}.}$$

The exponent changes at the target threshold $\beta = 1/2$.

Four mechanisms determine the loss curve

Combining the forcing pieces with Kesten's reduction,

$$\mathcal{P}(r, d) \asymp \underbrace{\mathcal{F}_{\text{pp}}(r)}_{\text{population bias}} + \underbrace{\mathcal{F}_{\text{ac}}(r, d)}_{\text{embedding bias}} + \underbrace{\mathcal{F}_0(d)}_{\text{model capacity}} + \underbrace{\frac{1}{\gamma B} \mathcal{K}_{\text{pp}}(r)}_{\text{stochastic variance}} .$$

For $\alpha > 1$,

$$\mathcal{K}_{\text{pp}}(r) \asymp r^{-2+1/\alpha} .$$

Compute optimality is now a finite competition among four explicit powers.

Each term has a different structural origin

term	scaling for $\alpha > 1$	structural meaning
$\mathcal{F}_{\text{pp}}$	$r^{-[1+(2\beta-1)/\alpha]}$	population modes relax
$\mathcal{F}_{\text{ac}}$	$d^{-1}r^{-1+1/\alpha}$	random features create a bulk
$\mathcal{F}_0$	$d^{-\alpha+\max(0,1-2\beta)}$	the feature span misses target mass
$\mathcal{K}_{\text{pp}}$	$(\gamma B)^{-1}r^{-2+1/\alpha}$	mini-batches continually inject variance

No term is universally dominant. The active pair changes with (α, β) .

A two-term balance sets the compute exponent

$$\mathcal{P}(r, d) \asymp r^{-p} + d^{-q}, \quad C \asymp dr.$$

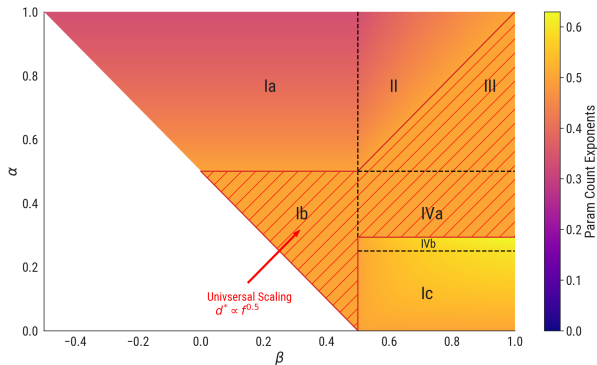
At fixed compute,

$$C^{-p} d^p \asymp d^{-q}.$$

Therefore

$$\boxed{\begin{aligned} d_{\star}(C) &\asymp C^{p/(p+q)}, \\ \mathcal{P}_{\star}(C) &\asymp C^{-pq/(p+q)}. \end{aligned}}$$

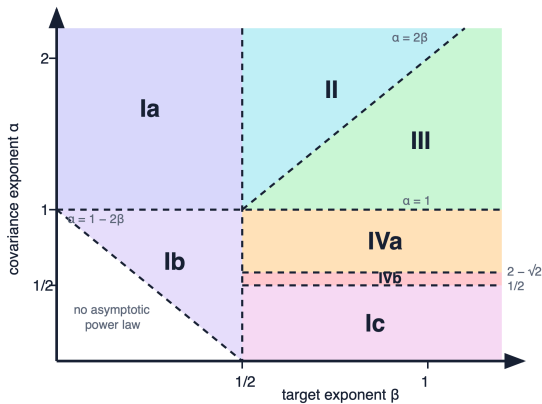
The calculation is simple only after identifying the correct active terms.



Red hatching: $d_{\star} \asymp C^{1/2}$. Source convention:

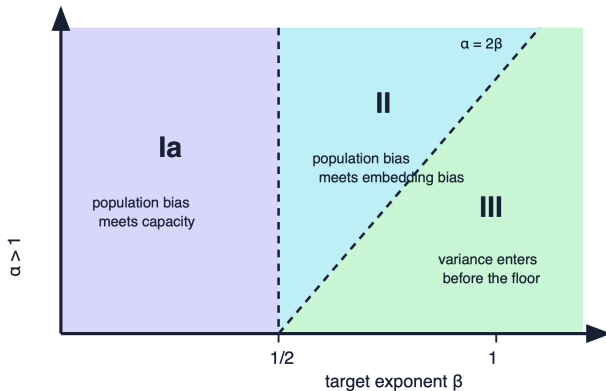
$\alpha_{\text{paper}} = \alpha_{\text{course}}/2$. Paquette et al. (2024), Appendix J, Fig. 8(a).

Four active terms produce seven scaling regions



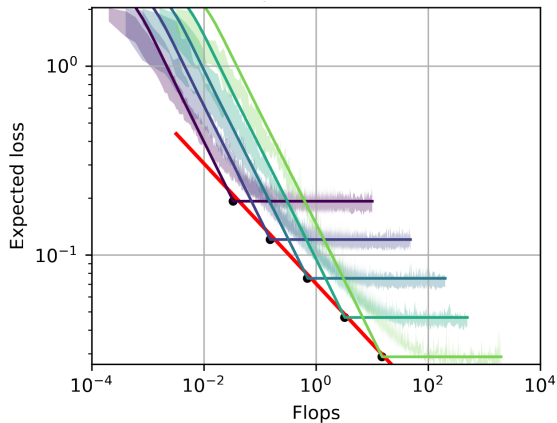
Course convention: $\lambda_j(\Sigma) \asymp j^{-\alpha}$.

Above the trace-class line, three balances remain



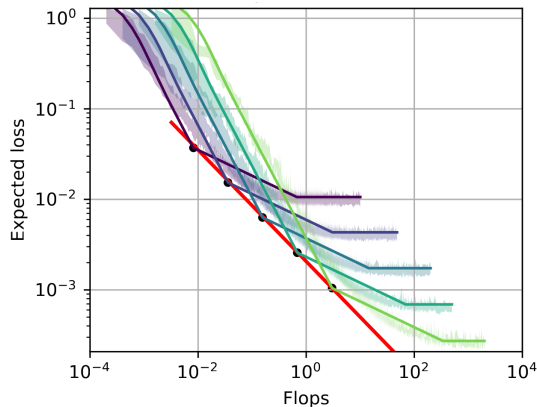
$\alpha = 2\beta$ separates embedding bias from variance-first behavior.

Phase Ia stops at the capacity floor



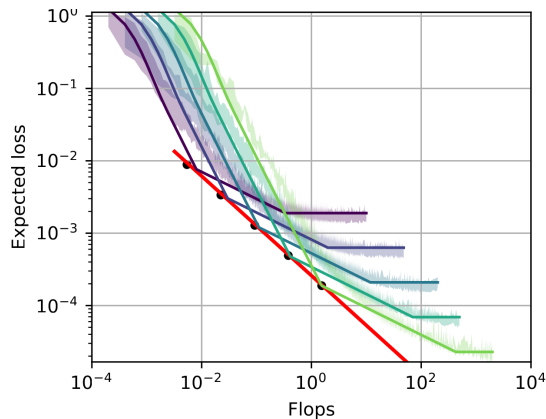
Course example: $\alpha = 1.2$, $\beta = 0.25$.

Phase II exposes embedding bias before the floor



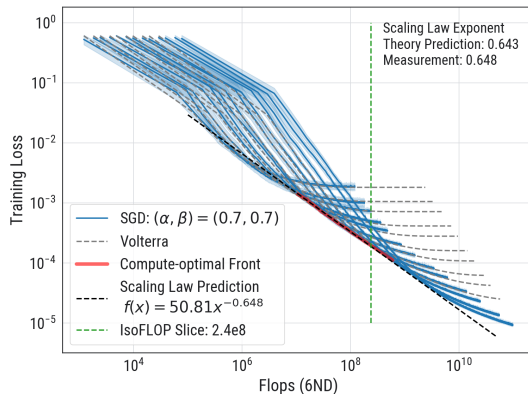
Course example: $\alpha = 1.4$, $\beta = 0.60$.

Phase III makes stochastic variance part of the frontier



Course example: $\alpha = 1.6$, $\beta = 0.90$.

The deterministic Volterra curve tracks SGD



Source legend $\alpha_{\text{source}} = 0.7$ corresponds to course $\alpha = 1.4$.

Selected references

- Elliot Paquette, Courtney Paquette, Lechao Xiao, and Jeffrey Pennington. “4+3 Phases of Compute-Optimal Neural Scaling Laws.” NeurIPS 2024. OpenReview
- Jordan Hoffmann et al. “Training Compute-Optimal Large Language Models.” NeurIPS 2022. arXiv:2203.15556
- Damien Ferbach, Katie Everett, Gauthier Gidel, Elliot Paquette, and Courtney Paquette. “Dimension-adapted Momentum Outscals SGD.” 2025. arXiv:2505.16098
- Alexander Maloney, Daniel A. Roberts, and James Sully. “A Solvable Model of Neural Scaling Laws.” 2022. arXiv:2210.16859